PROJECT TOPIC: AI Study Planner

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Assignment-07: Propose Solution to the Problem

1. Proposed Solution

To address the problem of static, manually maintained study timetables, we propose the AI Study Planner, a web-based tool that creates a study schedule for a student based on measurable inputs such as subjects, topics, exam dates, available study hours and topic difficulty. The main idea of this solution is not that the system "understands" the student in some intelligent or magical sense, but that it uses clear, measurable information, exam proximity, difficulty ratings, available time and quiz scores, to decide how much time should be spent on each topic. As the student's performance data changes, this decision is recalculated, and the schedule changes along with it.

2. System Workflow

From the point of view of a student using the system, the process looks like this. The student first enters their subjects, the topics under each subject, the exam date for each subject, how many hours they can study each day, and how difficult they personally find each topic. Based on this, the system calculates a priority for every topic and creates a study schedule that fits within the hours the student has available. The student then studies according to this schedule and attempts a short quiz after finishing a topic. The quiz result is used to update how well the student knows that topic, and the system recalculates the priorities and produces an updated timetable for the remaining days. Because of this, the timetable changes according to the student's actual progress instead of staying the same from start to end.

3. User Profile / Input

Each student maintains a simple profile within the system consisting of their subjects and the topics that fall under each subject, along with the exam date for each subject. Along with this, the student provides the number of hours they are realistically able to study per day or per week, and a difficulty rating for each topic. This information is entered once at the start and can be edited later if the student's plans change, for example if a new topic is added or an exam date is postponed.

4. Topic Priority Calculation

Every topic in the system is assigned a priority score, and this score decides how much attention that topic receives in the schedule. The score is based on three measurable factors: how close the
exam is, how difficult the student has rated the topic, and how weak the student's demonstrated performance is on that topic based on quiz scores. In simple terms:

Priority Score = Weighted Exam Urgency + Weighted Topic Difficulty + Weighted Student Weakness

None of these three factors is guessed by the system on its own. Exam urgency comes directly from the exam date entered by the student, difficulty comes from what the student has rated, and weakness comes from actual quiz scores recorded in the system. This keeps the calculation grounded in information the student has themselves provided or demonstrated, rather than in an assumption made by the system.

5. Time Allocation

Once priority scores are available for all topics under a subject, the available study hours are divided among the topics in proportion to their priority scores. A topic with a higher priority score receives a larger share of the available time, while a topic with a lower score receives less, although every topic is still given some minimum time so that no topic is skipped entirely. This allocation is done separately for each day leading up to the exam, so that the student always has a clear idea of what to study on a given day.

6. Quiz and Performance Feedback

After a topic has been studied according to the schedule, the system offers the student a short quiz related to that topic. This quiz is intentionally kept small so that it does not take up much of the student's time. The score from this quiz becomes the main source of feedback for the system, since it tells the system whether the student has actually understood the topic or whether more time and revision are still needed.

7. Adaptive Scheduling

Based on the quiz score, the weakness value for that topic is updated, which in turn changes its priority score. If the student scores poorly, the topic's priority increases and it receives more time in the next version of the schedule. If the student scores well, the priority decreases and the time is redirected towards other topics that need more attention. This is what makes the schedule adaptive: it is not fixed once at the beginning but is recalculated whenever new performance data is available.

8. Handling Missed Study Sessions

It is common for a student to be unable to follow the schedule exactly on a particular

day. To handle this, the system allows a student to mark a scheduled session as missed. The hours that were meant for that session are not simply discarded; instead, they are added back into the pool of hours to be redistributed over the remaining days before the exam, based on the same priority-based logic used elsewhere in the system. If too many sessions are missed and the remaining time is not enough to reasonably cover all topics before the exam, the system can flag this to the student so that they are aware of the shortfall rather than being caught by surprise close to the exam date.

9. Spaced Revision

Studying a topic once and moving on to the next one is often not enough for the student to retain it until the exam. To handle this, the system includes short revision slots for topics that were rated as difficult or on which the student initially performed poorly in the quiz, placing a brief review of these topics again closer to the exam date rather than only right after they were first studied. This is kept simple in this project, mainly using the difficulty rating and quiz performance to decide which topics deserve a revision slot, rather than a more advanced spaced-repetition algorithm.

10. Dashboard / Output

The output of the system is presented to the student through a simple dashboard. This includes a day-wise view of the study schedule, showing which topics are to be studied on which day and for how long, along with a list of topics sorted by their current priority so the student can see at a glance which areas need the most attention. The dashboard also shows a basic progress indicator for each subject, based on the topics completed and the quiz scores recorded so far, so the student has a clear picture of where they stand before the exam.

11. Expected Benefits

The main benefit of this system is that it removes the need for a student to manually decide, and repeatedly re-decide, how to divide limited study time among several subjects and topics. Because the schedule is based on exam urgency, difficulty and actual quiz performance rather than a one time guess, it is expected to give a more consistent allocation of study time to genuinely weaker topics than an unaided, manually prepared timetable usually achieves. It also reduces the effort involved in updating a timetable, since the student does not have to manually redraw their schedule every time their performance changes.

12. Limitations / Realistic Scope

This project does not claim to guarantee improved exam scores or grades, since exam performance also depends on many factors outside the scope of this system, such as the student's overall effort, the difficulty of the actual exam paper, and personal circumstances. The accuracy of the schedule also depends on how honestly and consistently the student rates topic difficulty and attempts the quizzes, since the system

has no way of verifying this information on its own. As this is a mini project, the priority calculation is kept rule-based rather than using a trained machine learning model, and features such as more advanced spaced repetition or automatic difficulty detection are left as possible future improvements rather than being part of the current scope.

13. Conclusion

The AI Study Planner proposed in this project offers a practical way for a student to move from a static, manually maintained study timetable to a schedule that adjusts itself based on exam urgency, topic difficulty and demonstrated performance. By using measurable inputs and a straightforward priority calculation, the system aims to make better use of a student's limited study time without requiring constant manual replanning, while remaining realistic about what such a tool can and cannot achieve.